Step-by-Step Setup Guide

How to Run the BlueEdge Patient Monitoring System on Another PC

EASY 5-MINUTE GUIDE

Overview: This project consists of two linked parts: **(1) Python Backend Gateway** (FastAPI on port `8000`) and **(2) Web Frontend Dashboard** (React + Vite on port `5173`). Follow the simple steps below on the new PC to get everything running smoothly.

1 Prerequisites Check (Install Once on the New PC)

Before extracting the project, verify that the new computer has **Python** and **Node.js** installed:

A. Python (3.10, 3.11, or 3.12)

1. Go to: **https://www.python.org/downloads/**
2. Download & run the installer.

> ⚠️ **CRITICAL:** Check the box **"Add python.exe to PATH"** at the bottom of the installer window before clicking Install!

B. Node.js (LTS version)

1. Go to: **https://nodejs.org/**
2. Download the **LTS version** (Recommended for Most Users).
3. Run installer and click *Next* → *Next* → *Install*.

2 Extract the Zip File on the New PC

1. Copy `Jigyasaa_project.zip` to the new PC (via Pen Drive, Google Drive, or Email).
2. Right-click on `Jigyasaa_project.zip` and choose **"Extract All..."**.
3. Select a folder destination (e.g. your **Desktop**) and click **Extract**.
4. Open the extracted folder (you will see `backend`, `frontend`, `run_backend.bat`, and `run_frontend.bat`).

3 Install Required Packages (One-Time Setup)

Open a Command Prompt or PowerShell inside the extracted project folder:

> 💡 **Shortcut:** Inside the extracted folder in Windows File Explorer, click in the address bar at the very top, type `cmd`, and press `Enter`.

Part A: Install Backend Python Libraries:

```
pip install -r requirements.txt
```

(This automatically installs FastAPI, Uvicorn, NumPy, WebSockets, and Pydantic)

Part B: Install Frontend Node.js Packages:

```
cd frontend
npm install
cd ..
```

(This automatically downloads the React & Tailwind UI packages)

Step-by-Step Setup Guide (Continued)

Running the Project & Verification

4 Run the Project (Every Time You Use It)

Method 1: Double-Click Launchers (Easiest & Recommended)

1. **Start Backend:** Double-click **run_backend.bat**.
   A black window will open showing: "*Uvicorn running on http://0.0.0.0:8000*".
   → Leave this window open in the background!
2. **Start Frontend:** Double-click **run_frontend.bat**.
   A second black window will open showing: "*Local: http://localhost:5173*".
   → Leave this window open as well!
3. **Open the Dashboard in Browser:** Open Google Chrome or Microsoft Edge and visit:
   👉 **http://localhost:5173**

Method 2: Using Terminal / Command Prompt

Terminal 1 (Backend)

```
cd backend
python main.py
```

Terminal 2 (Frontend)

```
cd frontend
npm run dev
```

5 How to Verify Everything Is Working Correctly

What you will see when it works:

- Visit **http://localhost:8000/docs** → You will see the interactive FastAPI documentation page.
- Visit **http://localhost:5173** → You will see the Live ICU Telemetry Dashboard with 12 clinical patient beds, real-time waveform visualizers, and AI risk prediction updates.

6 Quick Troubleshooting (Common Pitfalls)

- **'python' is not recognized:** Python was installed without checking "Add to PATH". Re-open the Python installer, select *Modify*, check *Add to PATH*, and restart your terminal.
- **'npm' is not recognized:** Close all terminals and restart the PC once so Windows reloads system environment paths.
- **Port 8000 or 5173 already in use:** Another process is already running on that port. Close all open command prompt windows and try again.
- **How to shut down servers when done:** Click on the terminal windows and press `Ctrl + C`, or simply close the command prompt windows.